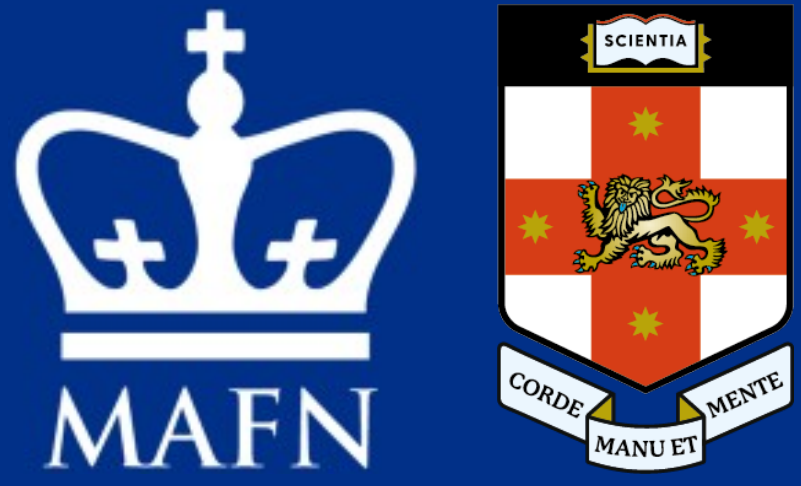




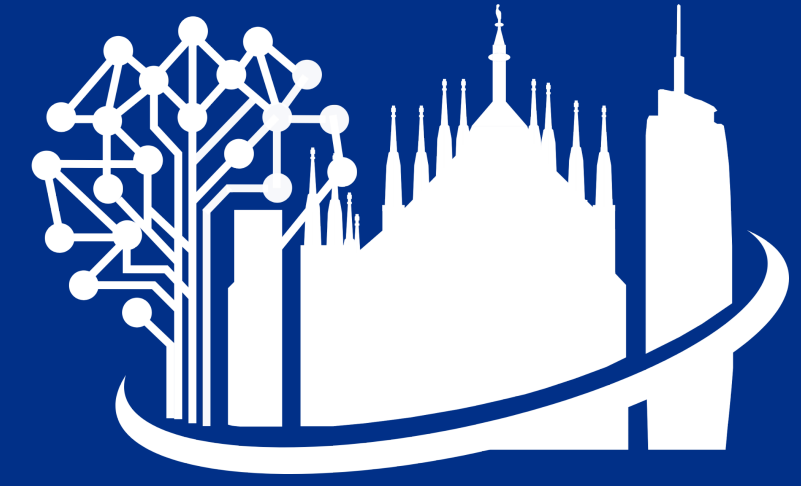
# Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

Nigel Li<sup>1,2</sup>

<sup>1</sup>Columbia University, New York

Yutong Han<sup>1</sup>

<sup>2</sup>University of New South Wales, Sydney



## Abstract

Reinforcement learning (RL) portfolio allocators are often given a tail-risk limit through a Lagrangian penalty on Conditional Value-at-Risk (CVaR). We show two ways in which this limit fails with no visible symptom, and we give a remedy for each.

| problem                                                                                                     | our solution                                                                                                          | result                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| The penalty never switches on, because the CVaR cost is on a far smaller scale than the reward (Section 3). | Plot $\lambda$ during training, then scale the dual step to the CVaR excess or standardise the cost signal.           | $\lambda$ rises from about 0 to between 0.1 and 0.3, and the penalty binds in training.                                             |
| Even a working penalty breaches the budget out of sample, in 75.1% of test weeks (Section 6).               | <b>A decision-time CVaR filter</b> that shrinks each over-budget trade toward minimum variance (Section 5).           | <b>Breach falls from 75.1% to 7.4%</b> , and CVaR <sub>0.99</sub> from 2.8% to 1.6%.                                                |
| One stress test ranks the learner above every optimiser, and a walk-forward reverses it (Section 4).        | Evaluate on a rolling walk-forward over two universes with Wilcoxon, Benjamini-Hochberg and Deflated Sharpe controls. | CVaR <sub>0.99</sub> <b>lower in 26 of 27 folds</b> ( $p = 8.8 \times 10^{-6}$ ). The Sharpe edge over optimisers does not survive. |

## 1 Motivation

In the week of 16 March 2020 the S&P 500 fell about 15% and the VIX closed at 82.69. An RL allocator trained only for return kept its equities. The same allocator trained with a CVaR budget had already moved into Treasuries and cash, and its drawdown was 7.7% against 16.3% (Fig. 1).

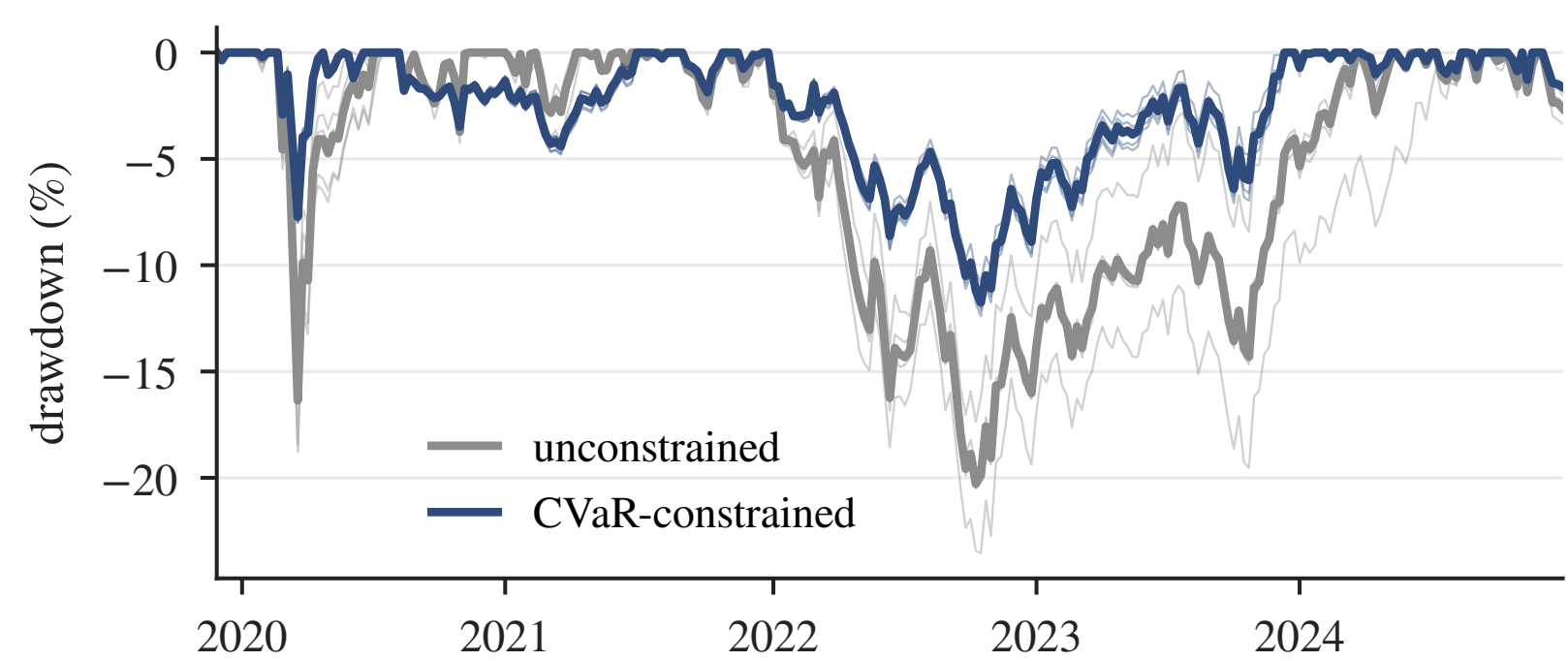


Figure 1. Drawdown over 2020 to 2024. Thick lines are five-seed means and thin lines individual seeds.

The budget is imposed with a Lagrangian penalty whose weight  $\lambda$  should grow while the portfolio is over budget [2].

## 2 Problem and measures

CVaR <sub>$\alpha$</sub>  is the average loss in the worst  $(1 - \alpha)$  share of weeks [1]. With  $\alpha = 0.95$  it averages the worst 5%, so it measures how large the losses are beyond the Value-at-Risk (VaR) threshold (Fig. 2).

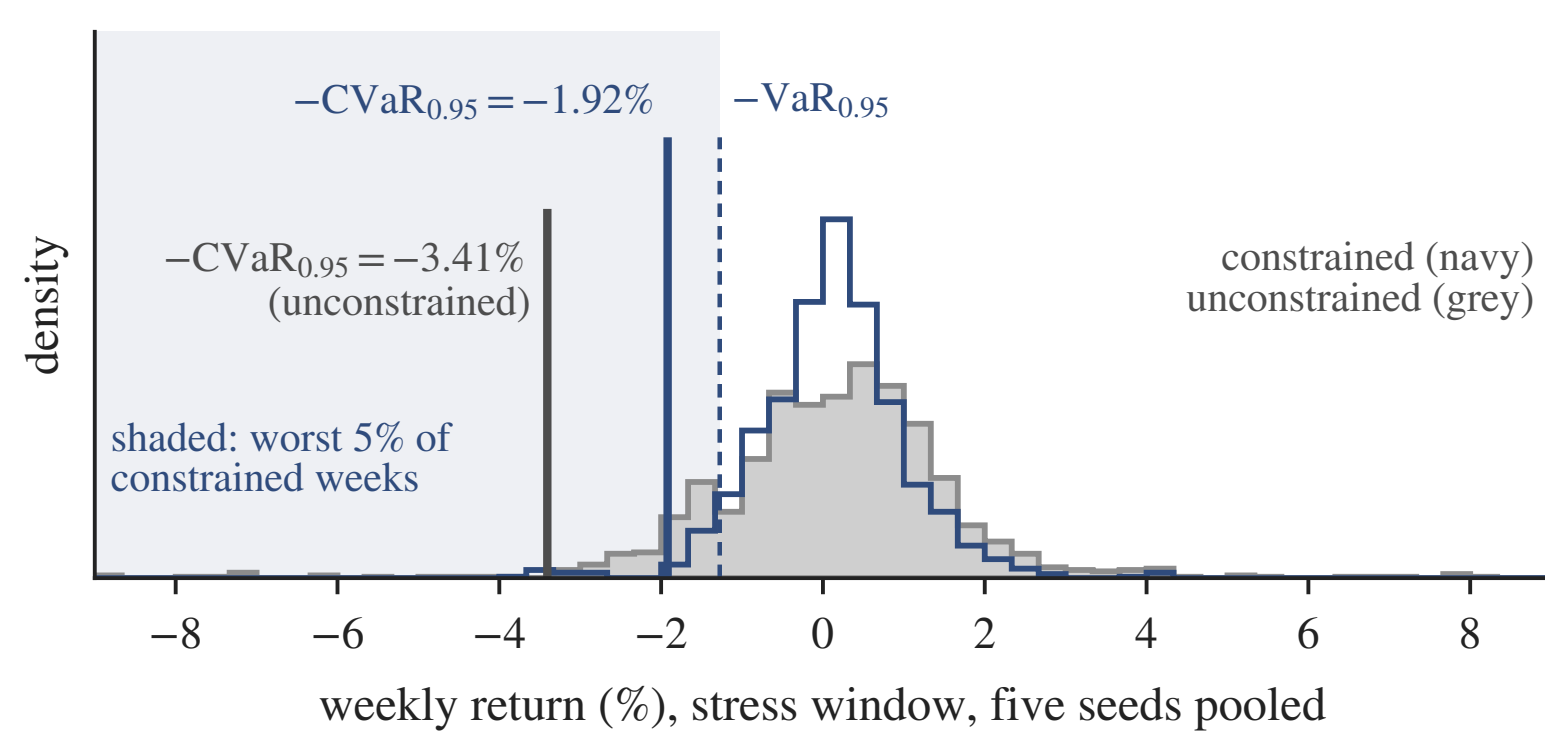


Figure 2. Weekly returns on the 2020 to 2024 stress window, five seeds pooled. The shaded left tail is the worst 5% of constrained weeks, and each line sits at minus that allocator's CVaR (per-seed means 1.89% and 3.33%).

Each week the allocator chooses long-only weights  $w_t$  on seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), using 887 weekly returns from 2008 to 2024. The problem is a constrained Markov decision process [2],

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad \text{CVaR}_{0.95}(L) \leq d = 1.2\%, \quad \text{CVaR}_{\alpha}(L) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}[(L - \zeta)^+] \right\}.$$

All measures use weekly net returns  $r_1, \dots, r_T$  and wealth  $W_t = \prod_{u \leq t} (1 + r_u)$ ,

$$\text{Sharpe} = \sqrt{52} \bar{r} / s_r,$$

$$\text{MDD} = \max_t (1 - W_t / \max_{u \leq t} W_u),$$

$$\widehat{\text{CVaR}}_{\alpha}(r) = -\text{mean}\{r_t : r_t \leq \hat{q}_{1-\alpha}(r)\},$$

$$B = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbf{1}[\widehat{\text{CVaR}}_{0.95}(r_{t-52:t-1}) > d],$$

where  $\hat{q}_{1-\alpha}$  is the empirical quantile and  $B$  is the breach rate over the test weeks  $\mathcal{T}$ , of which 5% may breach. Sharpe significance uses the Probabilistic and Deflated Sharpe ratios [4], with per-week  $\widehat{SR}$ , skewness  $\hat{\gamma}_3$ , kurtosis  $\hat{\gamma}_4$  and  $N$  configurations tried,

$$\text{PSR}(SR^*) = \Phi \left( \frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right), \quad SR_{\text{DSR}}^* = \hat{\sigma}_{SR} \left[ (1 - \gamma) \Phi^{-1} \left( 1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left( 1 - \frac{1}{N_e} \right) \right],$$

where  $\gamma$  is the Euler-Mascheroni constant.

## 3 Failure 1. The penalty weight never moves

With a dual step  $\eta_{\lambda} = 10^{-3}$ ,  $\lambda$  stayed near zero for all 1500 training updates (Fig. 3). The CVaR excess is about  $10^{-2}$  per week, so each update moved  $\lambda$  by about  $10^{-5}$ . With  $\eta_{\lambda} = 5.0$ ,  $\lambda$  rose to between 0.1 and 0.3. A model-free actor-critic failed in the same way when its cost signal (about  $10^{-3}$ ) was left unstandardised next to a unit-scale reward signal.

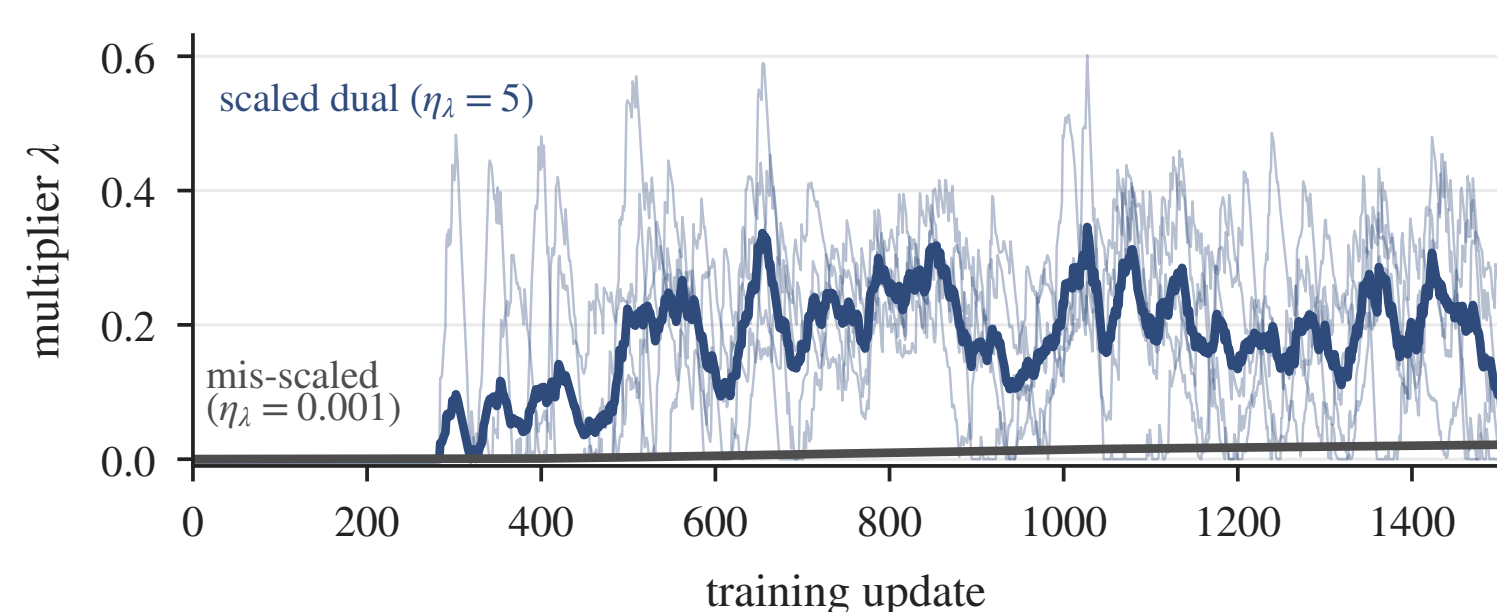


Figure 3. The penalty weight  $\lambda$  during training, five seeds.

Tail metrics do not reveal the failure. Even with  $\lambda$  near zero, the rest of the pipeline (the anchor, the projection and check-point selection under the CVaR limit) lowered CVaR<sub>99</sub> by 61% against the unconstrained learner (0.025 against 0.065, permutation test  $p = 0.0044$ ).

*Plot  $\lambda$  during training. A multiplier that stays at zero means the constraint is inactive.*

## 4 Failure 2. The test decides the ranking

A walk-forward test retrains the model on a rolling window and evaluates it on the next unseen block. On a 31-asset universe, one fixed stress split ranked the constrained learner above minimum variance (Sharpe 0.85 against 0.23). A 20-fold walk-forward reversed the order (0.74 against 1.40, Fig. 4). Ledoit-Wolf shrinkage [5] did not change the split result (0.25). The tail-risk reduction against unconstrained RL held under both tests.

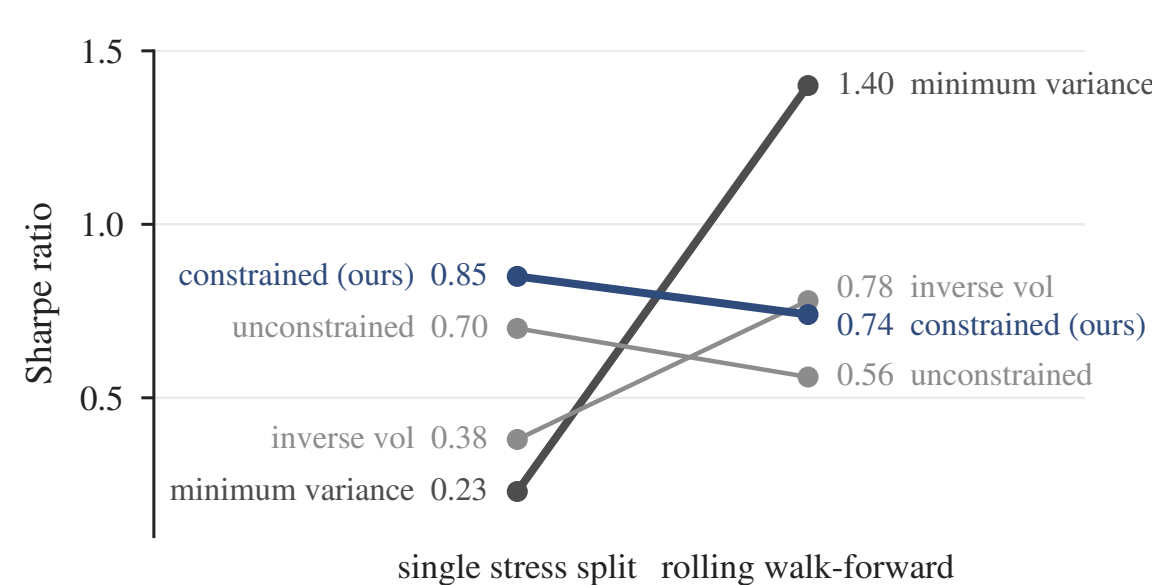


Figure 4. The same four methods ranked by two tests, 31-asset universe.

*Compare learners with optimisers on rolling walk-forward folds and report the Deflated Sharpe ratio [4].*

## 5 Our method

Our solution has two parts. The allocator is trained end to end with a CVaR penalty whose dual step matches the scale of the CVaR excess, and a filter checks every portfolio before it trades. Figure 5 writes the weekly decision as a chain of maps, and steps (i) to (iv) define each one.

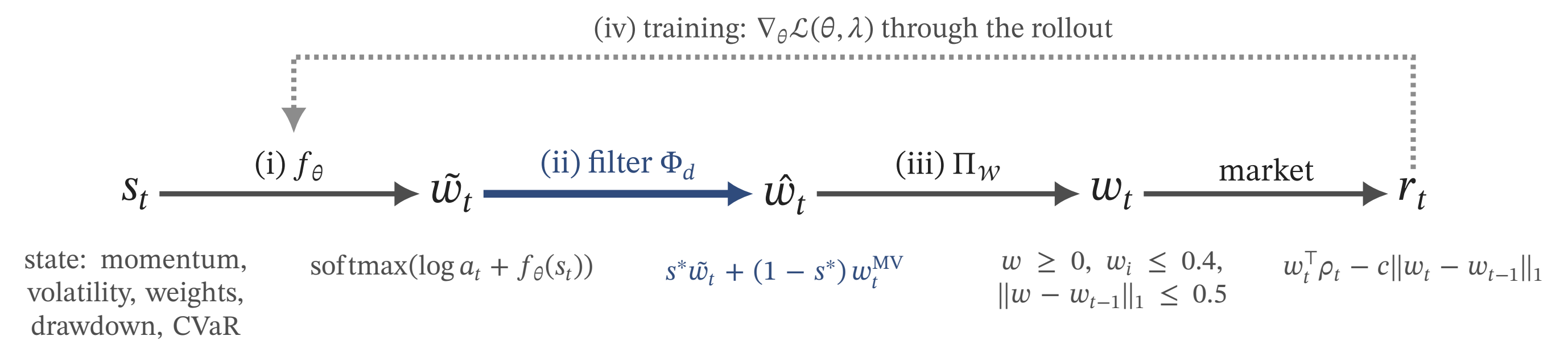


Figure 5. The decision at week  $t$  as a composition of maps.

(i) *Policy*. An MLP  $f_{\theta}$  tilts an adaptive inverse-volatility anchor  $a_t$ ,  $\tilde{w}_t = \text{softmax}(\log a_t + f_{\theta}(s_t))$ . At initialisation  $f_{\theta} \approx 0$ , so training starts from the anchor and learns a state-dependent tilt.

(ii) *Filter*. From the trailing 104 weekly returns it forecasts  $\widehat{\text{CVaR}}(w) = c_{\alpha} \sigma_p(w) - \mu_p(w)$ , with  $c_{\alpha} = \varphi(\Phi^{-1}(\alpha))/(1 - \alpha)$  and  $c_{0.95} = 2.06$ . An over-budget proposal is mixed with the trailing minimum-variance portfolio  $w_t^{\text{MV}}$ , keeping as much of the proposal as the budget allows,  $s^* = \max\{s \in [0, 1] : \widehat{\text{CVaR}}(s\tilde{w}_t + (1 - s)w_t^{\text{MV}}) \leq d\}$ . It is a safety layer in the sense of [3] and uses data up to week  $t - 1$  only.

(iii) *Projection*.  $\Pi_W$  enforces long-only weights,  $w_i \leq 0.40$  and  $\|w_t - w_{t-1}\|_1 \leq 0.50$ . The market then pays  $r_t$  net of a 5 bps cost per unit of turnover.

(iv) *Training*. On random windows of past weeks we back-propagate

$$\mathcal{L}(\theta, \lambda) = -\text{Sharpe}(r) + \lambda \text{ReLU}(\widehat{\text{CVaR}} - d) + \kappa \text{turnover}, \quad \lambda \leftarrow \max\{0, \lambda + \eta_{\lambda}(\widehat{\text{CVaR}} - d)\},$$

where  $\widehat{\text{CVaR}}$  is the mean of the worst 5% of window returns, so its gradient reaches only tail weeks. The kept check-point has the best validation objective among those within the validation CVaR limit.

Table 1. Training and evaluation details.

| component           | setting                                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| budget              | $\alpha = 0.95$ , $d = 1.2\%$ weekly CVaR, up to 5% of test weeks may breach                                                                                    |
| dual step           | $\eta_{\lambda} = 0.001$ (mis-scaled) or 5.0 (scaled)                                                                                                           |
| training            | 1500 updates per ablation arm and 2500 in the stress study, seeds 7, 13, 23, 42, 2025                                                                           |
| stress test         | train 2008 to 2018, test the 267 weeks from late 2019 to 2024                                                                                                   |
| walk-forward        | train 312, validate 52, test 26 weeks, step 26, 3-seed ensemble per fold                                                                                        |
| classical baselines | equal weight, inverse volatility, minimum variance, mean-variance, risk parity, min-CVaR LP, refit weekly on the trailing 104 weeks (52 for inverse volatility) |
| RL baselines        | A2C, PPO (36 configurations) and SAC (8), selected on validation Sharpe                                                                                         |
| statistics          | block bootstrap (2000 draws, block 4) with BCA intervals, Wilcoxon, Benjamini-Hochberg, PSR and DSR                                                             |

## 6 Fix. The filter keeps the portfolio within budget

A larger dual step made  $\lambda$  move, but the portfolio was then over budget in 75.1% of test weeks. Classical rules met the same budget in 92% of weeks, so the budget was attainable. Adding the filter lowered the breach rate to **7.4%** (Table 2). Raising  $\lambda$  during testing without the filter left it at 81.9%, so the improvement comes from the filter.

Table 2. Stress window, five-seed means. Breach is the share of test weeks over budget, with 5% tolerated.

| variant                                                        | Sharpe       | CVaR <sub>0.99</sub> | breach      |
|----------------------------------------------------------------|--------------|----------------------|-------------|
| no constraint                                                  | 0.626        | 6.5%                 | 100.0%      |
| penalty, $\eta_{\lambda} = 0.001$                              | 0.987        | 2.5%                 | 7.9%        |
| penalty, $\eta_{\lambda} = 5.0$                                | 0.911        | 2.8%                 | 75.1%       |
| penalty, $\lambda$ raised during testing                       | 0.826        | 3.1%                 | 81.9%       |
| <b>penalty, <math>\eta_{\lambda} = 5.0</math>, with filter</b> | <b>0.851</b> | <b>1.6%</b>          | <b>7.4%</b> |
| minimum variance                                               | 0.901        | 4.4%                 | 7.9%        |
| equal weight                                                   | 0.878        | 62.8%                |             |
| risk parity                                                    | 0.677        | 75.3%                |             |

## 7 Result. Lower tail risk in every test

The walk-forward test retrains every 26 weeks on the previous six years and tests on the next 26 unseen weeks, giving 20 folds per universe from 2015 to 2024. In Figure 6 each fold is one point. Below the diagonal the constraint lowered CVaR<sub>0.99</sub>. On it the budget never bound, so both models held the same portfolio.

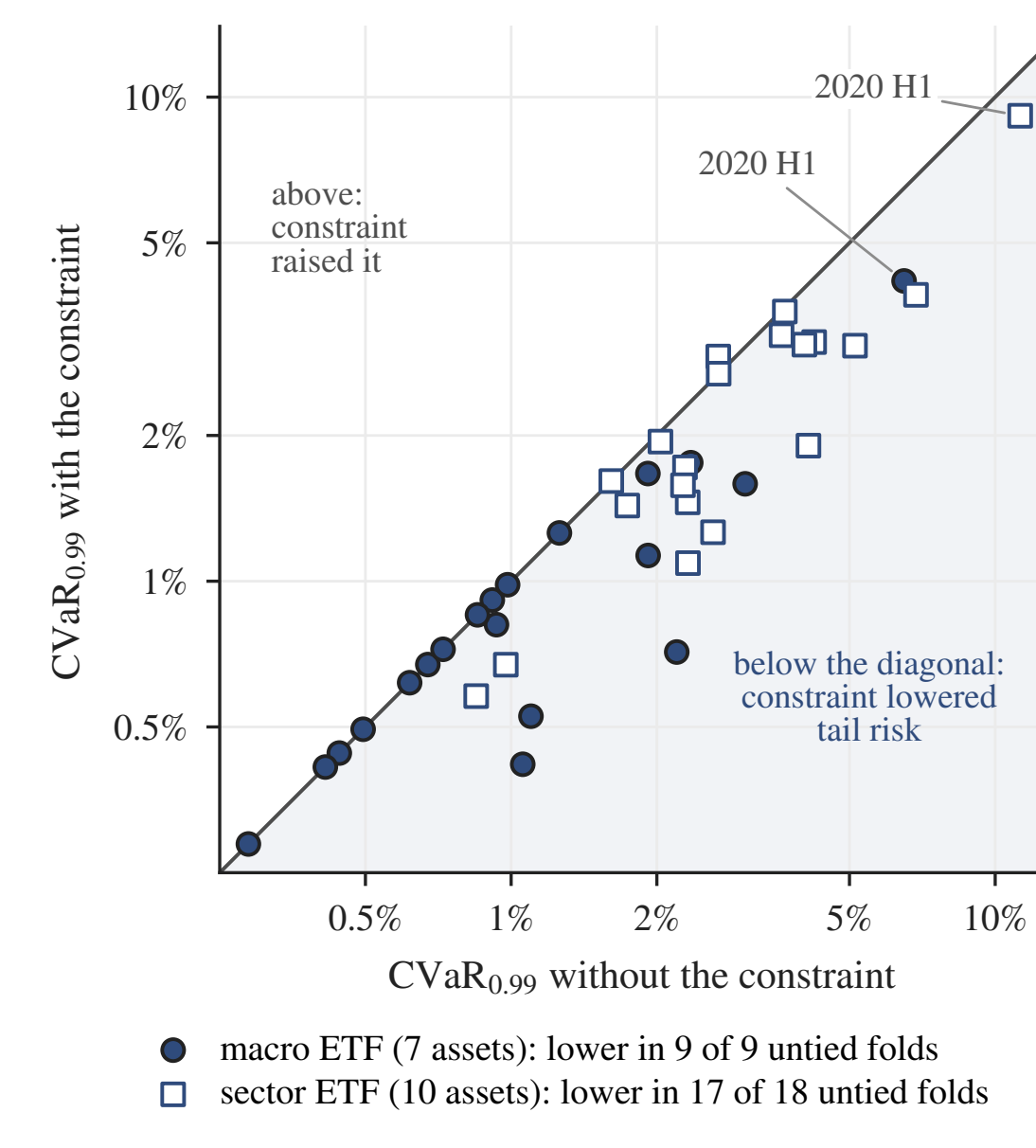


Figure 6. Walk-forward folds, CVaR<sub>0.99</sub> without against with the constraint (log scale). The largest reductions came in the first half of 2020.

Table 3. Tail risk with and without the constraint in every test setting.

| setting                       | test periods       | CVaR <sub>0.99</sub> without $\rightarrow$ with | evidence                                                      |
|-------------------------------|--------------------|-------------------------------------------------|---------------------------------------------------------------|
| stress window, 7 ETFs         | 267 weeks, 5 seeds | 6.47% $\rightarrow$ 3.27%                       | 95% BCa CI $[-3.5, -2.2]$ pp                                  |
| stress window, 10 sector ETFs | 267 weeks, 3 seeds | 10.9% $\rightarrow$ 7.6%                        | 95% BCa CI $[-5.0, -3.2]$ pp                                  |
| walk-forward, 7 ETFs          | 20 folds           | 1.43% $\rightarrow$ 1.02%                       | lower in 9 of 9, $p = 0.0077$                                 |
| walk-forward, 10 sector ETFs  | 20 folds           | 3.36% $\rightarrow$ 2.50%                       | lower in 17 of 18, $p = 0.0003$                               |
| <b>both universes pooled</b>  | <b>40 folds</b>    |                                                 | <b>lower in 26 of 27, <math>p = 8.8 \times 10^{-6}</math></b> |
| walk-forward, 31 assets       | 20 folds           |                                                 | $p = 6 \times 10^{-4}$                                        |

Fold counts exclude tied folds. Each  $p$  is a Wilcoxon signed-rank test on  $\Delta_k = \text{CVaR}_{0.99,k}^{\text{con}} - \text{CVaR}_{0.99,k}^{\text{unc}}$ , with  $W^+ = \sum_{k: \Delta_k \neq 0} R_k \mathbf{1}[\Delta_k > 0]$  and  $R_k$  the rank of  $|\Delta_k|$ . Both universes pass Benjamini-Hochberg [6]. Calendar-clustered  $p = 3.3 \times 10^{-4}$ . PSR(0)  $\geq 0.99$  in both, but the Sharpe edge over classical baselines did not survive DSR.

## 8 Limitations and future work

The learned allocator did not beat rolling minimum variance on Sharpe out of sample, and its Sharpe advantage over classical baselines did not survive the Deflated Sharpe ratio. The study uses weekly data with proportional costs and no market impact, the protocol reversal was shown on one 31-asset window, and the filter's Gaussian forecast can understate fat-tailed losses.

The next steps are a fat-tailed CVaR forecast inside the filter, daily data with market-impact costs, and the protocol comparison on more universes.

**References.** [1] R. T. Rockafellar, S. Uryasev, Optimization of conditional value-at-risk. *J. Risk* 2 (2000). [2] E. Altman. *Constrained Markov Decision Processes*. 1999. [3] G. Dalal et al. Safe exploration in continuous action spaces. arXiv:1801.08757, 2018. [4] D. H. Bailey, M. López de Prado. The deflated Sharpe ratio. *JPM* 40 (2014). [5] O. Ledoit, M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *JMVA* 88 (2004). [6] Y. Benjamini, Y. Hochberg. Controlling the false discovery rate. *JRSS B* 57 (1995).  
 ✉ nl2992@columbia.edu nigel.li@unsw.edu.au yh3915@columbia.edu  
 GitHub.com/nl2992/ICAIF\_cvar-rl-portfolio-allocator

